

Micro Energy Harvesting

Formulary

1. Thermoelectric Energy Harvester

- Thermoelectric figure of merit ZT of a material

$$ZT = \frac{S^2 \cdot \sigma}{\kappa} \cdot T$$

$$S : \text{Seebeck - coefficient} \quad \left[\frac{\text{V}}{\text{K}} \right]$$

$$\kappa : \text{Thermal conductivity} \quad \left[\frac{\text{W}}{\text{m} \cdot \text{K}} \right]$$

$$\sigma : \text{Electrical conductivity} \quad \left[\frac{1}{\Omega \cdot \text{m}} \right]$$

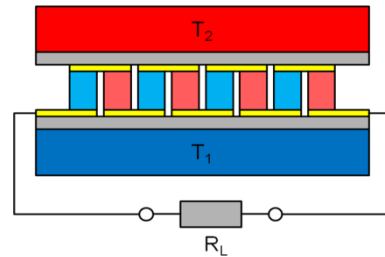
$$T : \text{Temperature} \quad [\text{K}]$$

- Thermoelectric efficiency

$$\eta_c = \frac{T_2 - T_1}{T_2}$$

T_2 : Temperature of the hot side

T_1 : Temperature of the cold side



- Thermogenerator: Seebeck-voltage/current

$$U_g = \alpha_g \cdot (T_h - T_c); \quad I_L = \frac{U_g}{R_g + R_L};$$

U_g : Generator voltage

I_L : Current flowing through generator and load resistor

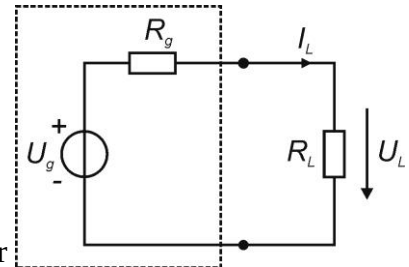
$T_{h,c}$: Temperature of hot/cold side

α_g : Seebeck - coefficient of the generator ($\alpha_g = m \cdot (S_p - S_n)$)

with m thermocouples

R_g : Internal resistance of the generator

R_L : Load resistance



- Thermal resistance of the generator with m thermocouples

$$R_{th,gen} = \frac{1}{2m} \cdot \frac{l}{\lambda \cdot A_{TB}} \quad \left[\frac{\text{K}}{\text{W}} \right]$$

l – Length of one thermo - leg

λ – thermal conductivity

A_{TB} – Area of one thermo - leg

- Electrical resistance of the generator with m thermocouples

$$R_{el,gen} = R_g = 2m \cdot \frac{\rho \cdot l}{A_{TB}} \quad [\Omega]$$

ρ – Specific electrical resistance $[\Omega m]$

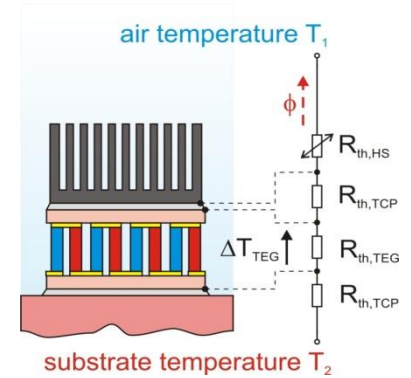
- Thermoelectric figure of merit ZT of the generator

$$ZT_{gen} = \frac{\alpha_{gen}^2 \cdot R_{th,gen}}{R_{el,gen}} \cdot T$$

- Thermal resistances at the generator

$$\Delta T_g = T_h - T_c = \left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{th,hot} + R_{th,cold}} \right) \cdot (T_2 - T_1)$$

Nomenclature in the picture : $\Delta T_g \equiv \Delta T_{TEG}$; $K_g \equiv R_{th,TEG}$



- Total output of the system (th. & el. voltage divider)

$$P_{el} = I_L^2 \cdot R_L = \alpha_g^2 (T_2 - T_1)^2 \cdot \left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{th,hot} + R_{th,cold}} \right)^2 \cdot \frac{R_L}{(R_g + R_L)^2}$$

- Total output of the system with DC/DC step-up converter

$$P_{el,DCDC} = I_L^2 \cdot R_L \cdot \eta_{DC/DC} = \alpha_g^2 (T_2 - T_1)^2 \cdot \left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{th,hot} + R_{th,cold}} \right)^2 \cdot \frac{R_L}{(R_g + R_L)^2} \cdot \eta_{DC/DC}$$

2. Undamped harmonic oscillator

- General DE of the undamped harmonic oscillator:

$$m \frac{\partial^2}{\partial z^2} + kz = 0$$

3. Damped harmonic oscillator

- General DE of the damped harmonic oscillator:

$$m\ddot{z} + b_m\dot{z} + kz = 0$$

- General solution of the equation of motion:

$$z(t) = A_0 \cdot \cos(\omega_0 \cdot t)$$

- Damping coefficient

$$b_m = \delta \cdot 2m$$

- Quality factor

$$Q = \frac{1}{2\zeta_m} = \frac{m\omega_0}{b_m}$$

- Lehr's damping measure / dimensionless damping coefficient

$$\zeta_m = \frac{\delta}{\omega_0} = \frac{b_m}{2m\omega_0}$$

- Damped resonance Frequency

$$\omega_d = \omega_0 \sqrt{1 - \zeta_m^2}$$

4. General equation for vibration harvesters

- Vibration amplitude Z of the seismic mass in dependency of the excitation vibration Y of the system

$$|Z| = \frac{\omega_{exc}^2 \cdot |Y|}{\sqrt{(\omega_0^2 - \omega_{exc}^2)^2 + [2 \cdot (\zeta_m + \zeta_{el}) \cdot \omega_0 \cdot \omega_{exc}]^2}}$$

- Maximum harvestable electrical power

$$P_{el}(t) = b_{el} \cdot \dot{z}(t)^2 = 2 \cdot \zeta_{el} \cdot m \cdot \omega_0 \cdot \dot{z}(t)^2$$

- Averaged electrical power for constant external excitation ω_{exc} with **acceleration amplitude A_g**

$$P_{el,av}(\omega_{exc}) = \frac{m \cdot \zeta_{el} \cdot \left(\frac{\omega_{exc}}{\omega_0}\right)^3 \cdot \omega_{exc}^3}{\left[1 - \left(\frac{\omega_{exc}}{\omega_0}\right)^2\right]^2 + 4 \cdot (\zeta_m + \zeta_{el})^2 \cdot \left(\frac{\omega_{exc}}{\omega_0}\right)^2} \cdot A_g^2$$

- Maximum electrical power for a constant external excitation in resonance $\omega_{exc} = \omega_0$ with **deflection amplitude A_0**

$$P_{el,max}(\omega_{exc}) = \frac{m \cdot \zeta_{el}}{4 \cdot \omega_0 (\zeta_m + \zeta_{el})^2} \cdot A_0^2$$

- Conversion between **acceleration amplitude A_g** und **deflection amplitude A_0**

$$A_g = A_0 \omega_0^2 \left[m / s^2 \right]$$

5. Piezoelectric Harvester

- Electrical damping and optimal load resistance
(k – piezoelectric coupling factor (mostly $\cong 0,15$); R – load resistance;
 C_p – capacity of the generator)

$$\zeta_{el}(s) = \frac{\omega \cdot k^2}{2 \cdot \left(s + \frac{1}{R \cdot C_p}\right)^2} \stackrel{!}{=} \zeta_m(s) \Rightarrow R_{opt} = \frac{1}{\omega C} \cdot \frac{2 \cdot \zeta_m}{\sqrt{4 \cdot \zeta_m^2 + k^4}}$$

- Output power of a piezoelectric bimorph-generator, in case that the piezo beams are connected in series

$$P = \frac{1}{\omega_{res}^2} \frac{R \cdot C_p^2 \cdot \left(\frac{E_c \cdot d_{31} \cdot t_c \cdot b^*}{1 \cdot \epsilon_c} \right)^2}{(4 \cdot \zeta_m^2 + k^4)(R \cdot C_p \cdot \omega_{res})^2 + 4 \cdot \zeta_m \cdot k^2 (R \cdot C_p \cdot \omega_{res}) + 2 \zeta_m^2} \cdot A_g^2$$

- With the geometric constants b^* and b^{**}

$$b^* = \frac{3b \cdot (2l_b + l_m - l_e)}{l_b^2 \cdot (2l_b + \frac{3}{2}l_m)} \quad b^{**} = \frac{2I_{eff}}{b \cdot (2l_b + l_m - l_e)}$$

- Effective geometric moment of inertia

$$I_{eff} = 2 \left[\frac{w \cdot t_c^3}{12} + w \cdot t_c \cdot b^2 \right] + \frac{w \cdot t_s^3}{12} \cdot \frac{E_c}{E_s}$$

E_c = Youngs modulus of piezo

E_s = Youngs modulus of substrate

- Resonance frequency of the bimorph beam

$$\omega_{res} = \sqrt{\frac{E_c}{m} \cdot \frac{6}{l_b^2 \cdot (2l_b + \frac{3}{2}l_m)} \cdot I_{eff}}$$

t_c : Thickness of piezo

t_s : Thickness of substrate

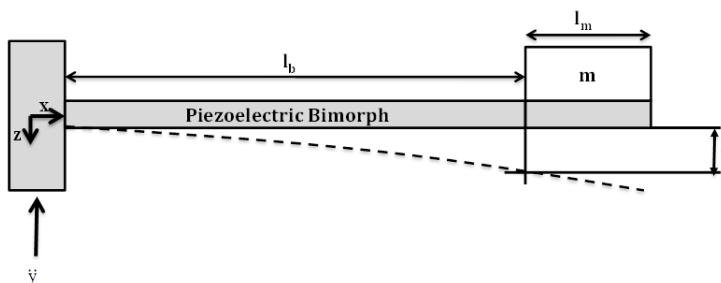
b : Distance between the center lines of the individual layers

w : beam width

l_b : Beam length

l_e : Length of electrode = Beam length

l_m : Length of the seismic mass



6. Electromagnetic Harvester

- Induced voltage

$$U = -N \cdot \frac{d\Phi}{dt}$$

- Electromagnetic coupling factor

$$k = \frac{U}{\dot{z}} \quad \begin{array}{l} U - \text{Induced voltage} \\ \dot{z} - \text{velocity} \end{array}$$

$$k = N \cdot A \cdot \left. \frac{dB}{dz} \right|_{z_0} \approx \text{const}$$

- Electrical damping

$$\zeta_{el}(s) = \frac{k^2}{m \cdot \omega_0} \cdot \frac{1}{(R_0 + R_L) + s \cdot L}$$

R_0 : Coil resistance

R_L : Load resistance

L : Inductivity of the coil

k_{mag} : Electromagnetic coupling coefficient

- Optimal load resistance

$$R_{L,opt} = \frac{k^2}{m \cdot \omega_0 \cdot \zeta_m} + R_0$$

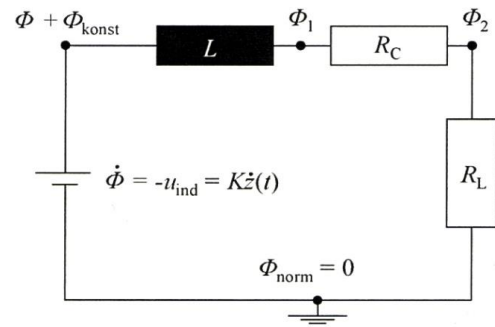
ζ_m : Mechanical damping

- Output power at the load resistance

$$P_{el} \cdot \eta = \frac{R_L \cdot k_{mag}^2 \cdot m^2}{4[k^2 + 2 \cdot m \cdot \omega_0 \cdot \zeta_m \cdot (R_0 + R_L)]^2} \cdot A_g^2 \cdot \frac{R_L}{R_L + R_0}$$

- Conversion between **acceleration amplitude A_g** und **deflection amplitude A_0**

$$A_g = A_0 \omega_0^2 \left[\frac{m}{s^2} \right]$$



7. Photovoltaic Harvester

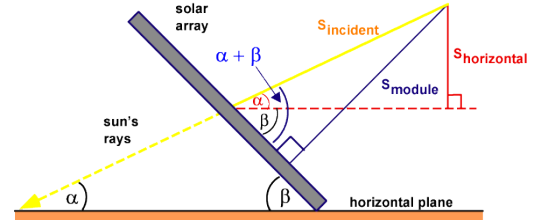
- Air-Mass

$$AM = \frac{1}{\cos(\theta) + 0.50572 \cdot (96.07995 - \theta)^{-1.6364}}$$

θ : angle of zenith (48.2° for AM 1.5)

- Radiation for modules tilted directly towards the equator

$$\begin{aligned} S_h &= S_i \cdot \sin \alpha \\ S_m &= S_i \cdot \sin(\alpha + \beta) \\ \Rightarrow S_m &= \frac{S_h \cdot \sin(\alpha + \beta)}{\sin \alpha} \end{aligned}$$



for arbitrary module orientation:

with ψ and θ : Azimuth angles of module and sun

$$S_m = S_i \cdot [\cos \alpha \sin \beta \cos(\psi - \theta) + \sin \alpha \cos \beta]$$

- Spectral Response

$$SR = \frac{q\lambda}{hc} QE \text{ in A/W}$$

- IV-Curve of solar cells

$$I = \underbrace{\tilde{I}_L}_{\text{light induced current}} - \underbrace{I_0 \cdot \left(e^{\frac{qV}{nkT}} - 1 \right)}_{\text{dark current}}$$

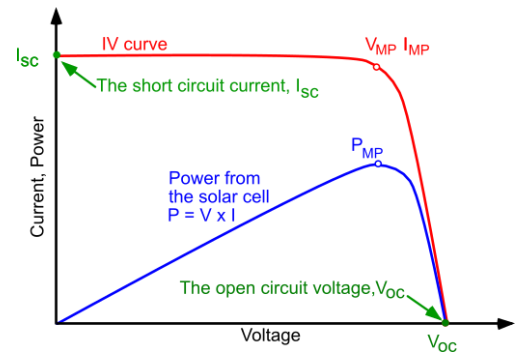
q : elementary charge

V : voltage

n : ideality factor

k : Boltzmann-constant

T : temperature



Short Circuit Current I_{SC}

$$I_{SC} \approx I_L$$

Open Circuit Voltage V_{OC}

$$V_{OC} = \frac{nkT}{q} \ln \left(\frac{I_L}{I_0} + 1 \right)$$

Fill Factor FF

$$FF = \frac{P_{MP}}{V_{OC} \cdot I_{OC}} = \frac{V_{MP} \cdot I_{MP}}{V_{OC} \cdot I_{OC}}$$

V_{MP}, I_{MP} : Maximum Power Point (MP) voltage and current

- Efficiency

$$\eta = \frac{P_{max}}{P_{in}} = \frac{V_{OC} \cdot I_{SC} \cdot FF}{P_{in}}$$

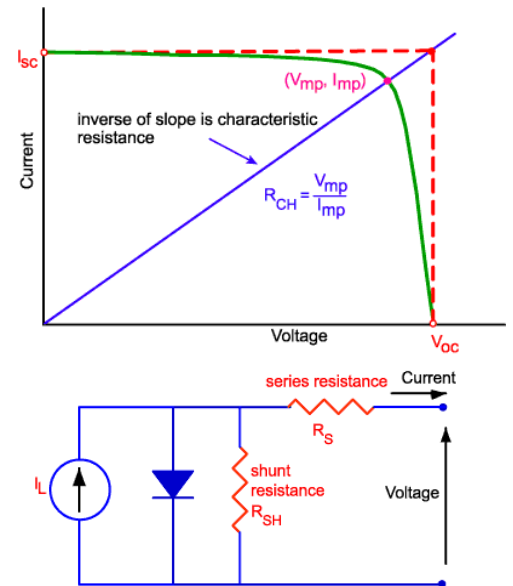
- Resistances

Characteristic resistance

$$R_{CH} = \frac{V_{MP}}{I_{MP}} \approx \frac{V_{OC}}{I_{SC}}$$

Series Resistance R_S

Shunt Resistance R_{SH}

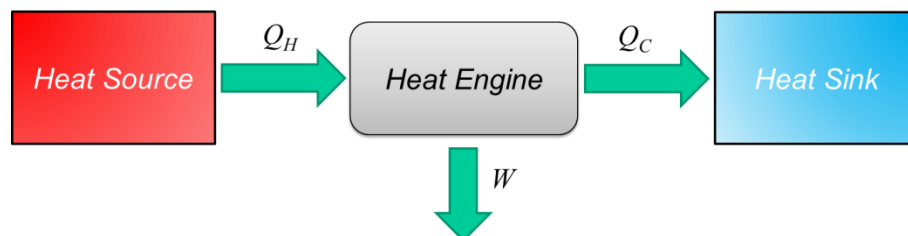


Influence on Fill Factor:

$$FF = FF_0 \cdot \left\{ (1 - 1.1r_S) + \frac{r_S^2}{5.4} \right\} \left\{ 1 - \frac{V_{OC} + 0.7}{V_{OC}} \cdot \frac{FF_0}{r_{SH}} \cdot \left[(1 - 1.1r_S) + \frac{r_S^2}{5.4} \right] \right\}$$

$$\text{with } r_S = \frac{R_S}{R_{CH}} \text{ and } r_{SH} = \frac{R_{SH}}{R_{CH}}$$

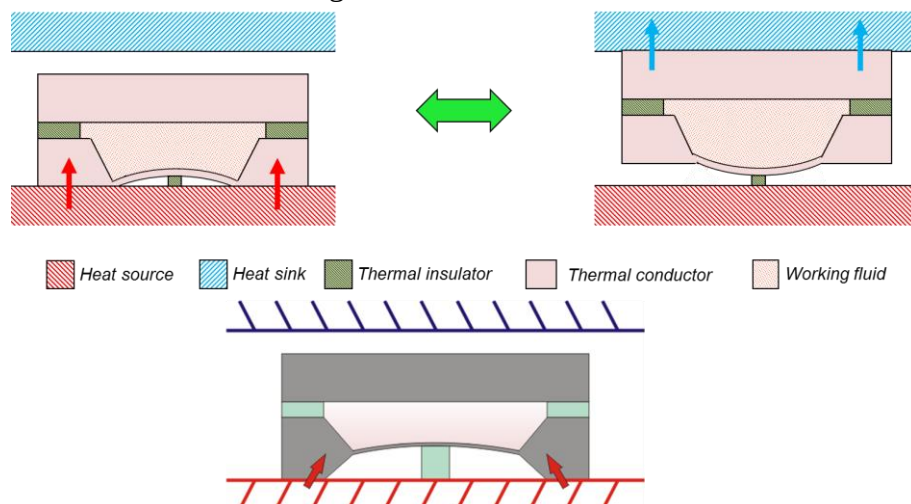
8. Thermomechanical Energy Harvester



- Efficiency generally:

$$\eta = \frac{Q_H - Q_C}{Q_H}$$

- IMTEK – Micro Heat engine



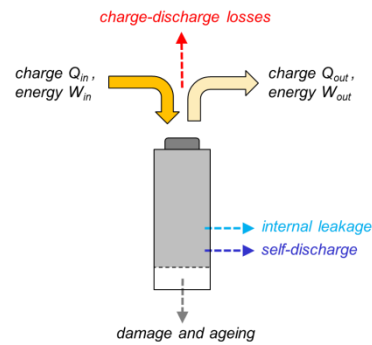
9. Power Management

- Coulomb efficiency:

$$\eta_C = \frac{Q_{out}}{Q_{in}}$$

- Energy efficiency:

$$\eta_E = \frac{E_{out}}{E_{in}} = \frac{U_{dis} \cdot I_{dis} \cdot t}{U_{chg} \cdot I_{chg} \cdot t}$$



- Self-Discharge-Rate

$$SDR = \frac{\Delta C}{C_0 \cdot \Delta t}$$

C_0 : initial capacity [As]

ΔC : loss of capacity over Δt [As]

Δt : test interval [s]

- Multilayer ceramic chip capacitor (MLCC)
general equation for one layer

$$C = \epsilon_0 \cdot \epsilon_r \cdot \frac{A}{d} \quad // \quad R_p = \rho \cdot \frac{d}{A}$$

- Equivalent Series Resistance – ESR

$$R_{ESR} \approx \frac{R_p \cdot R_s}{R_p + R_s}$$

- Loss factor

$$\tan(\delta) = \frac{\text{Re}|Z|}{\text{Im}|Z|} = R_{ESR} \cdot \omega C$$

- Self-discharge time

$$\tau = R_p \cdot C$$

- Time period until 50 % of the energy is lost

$$t_{50} = 0.35 \cdot \tau$$

- Continuous power dissipation

$$P_{loss}(t) = \frac{U_C^2(t)}{R_p}$$

